

Supplementary Information: IRSpectra-Bench — protocol, scale tables, cases, and reproducibility

Anonymous authors

Companion to the ICLR 2027 submission

Generation Recall, Not Verification, Binds LLM Structure Elucidation
from Literature Spectra

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This SI holds volume that does not belong on the 9-page ICLR main text. Every accuracy number is copied from a committed lock (docs/POOLED_HEADLINE_2026-09-16.md) or from the 2026-09-20 night score note (docs/NIGHT_POOL_2026-09-20.md). No CIs are invented for expand-500. No forward-verify precision is claimed on the 106 or the 230. The paper headline remains $n=300$. Figure and listing numbers below refer to the main PDF (Fig. 1 = protocol + $n=194$ wall; Listing 1 = IRexp payload). They are not re-numbered here.

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1 How to read this SI (figure / listing map)

The night main-text plate and listing are the source of truth. This SI does not fight that numbering.

Table 1: Main-text floats vs this SI. Fig. 1 is the Figma Bro lead plate.

main label	object	SI role
Fig. fig1	Fig. 1 (fig1_lead_overview)	Figma Bro lead: input $\rightarrow$ generate $\rightarrow$ verify, factorisation, and $n=194$ wall 58/7/129. Wall is not a pooled $n=300$ diagnosis.
Fig. fig-chemspace	Fig. 2	$n=300$ chemistry; panels on validate-clean $n=295$.
Listing lst:payload	Listing 1 (listing1_irexp_record)	IReXP JSON+molecule (NTJIYHYWVZY-BEX; $C_{12}H_{16}N_2O_2$). SMILES/InChIKey are resource fields, not solver payload.
Fig. fig-wall	appendix (fig_wall_diagnostic)	Repeat of the Fig. 1 wall on the locked $n=194$ slice only.
Fig. fig-robustness	Fig. 4	Formula-only / recency / four-vendor.
Fig. fig-mechanism	Fig. 5	Locked v3-R25 verify-save. Reused in §9.
Fig. fig3-method	Fig. 6	Inference ladder on the 60-compound arm.
App. Tables 7–10	cases / +106 / $n=295$ / scale	Same integers as §5–7.
SI Figs. S1–S3	reprints	Same three files as main Fig. 1 / Listing 1 / appendix wall (§2). Not a second lead figure.

Equations in the main text are Eqs. (1)–(7): factorisation, InChIKey-14 indicator, top-1, recall@ k , prec.|recall, ^{13}C chamfer, pooled estimator. They are restated in §4 with the same meaning. No new metric is defined.

2 Visual plates (reprints; not a second Figure 1)

The files below are the same as the main PDF. They are reprinted so this SI is readable alone. Main-text Figure 1 (fig1_lead_overview) remains the only lead diagram. Panel letters (a)/(b)/(c) from an earlier draft do not apply: the Figma Bro plate is one figure (protocol strip, factorisation, wall).

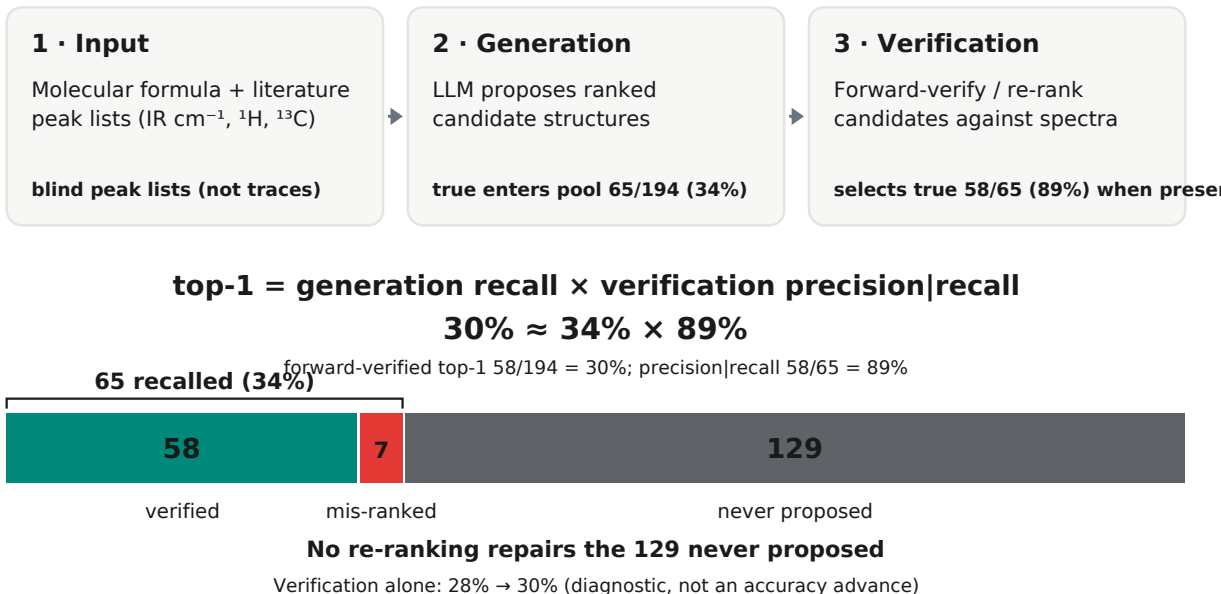


Figure S1: Reprint of main-text Figure 1. Input $\rightarrow$ generate $\rightarrow$ verify; 30% $\approx$ 34% $\times$ 89%; locked $n=194$ wall 58/7/129. Not a pooled $n=300$ wall.

3 Blind protocol and pre-registration

3.1 Solver-facing contract

Each IRSpectra-Bench item is a four-field JSON object. The solver sees only

$\{\text{qid}, \text{formula}, \text{ir_bands_cm-1}, \text{h_nmr}, \text{c_nmr}\}$.

No name, SMILES, InChIKey, scaffold, atom-type hint, retrieved spectrum, licence stamp, or PMC id enters the prompt (main-text Listing 1, adapted from one IReXP record). Gold constitution is scored offline by the InChIKey-14 indicator (§4).

Difficulty is declared a priori by RDKit ring analysis: simple iff ≤ 2 rings, no fused/spiro/bridge-head system, ≤ 22 heavy atoms; else complex. The released $n=300$ bench is 50/50 by design

Listing 1. Example IReXP JSON record (commercial DoR).

```
{
  "id": "NTJIYHYWVZYBEX-UHFFFAOYSA-N",
  "ir_bands_cm-1": [3337.0, 2969.0, 1633.0, 1556.0,
    1505.0, 1240.0, 1181.0, 1071.0, 826.0],
  "ir_source": "experimental",
  "source_doi": "PMC:13029360",
  "pmcid": "PMC13029360",
  "h_nmr": "8.07 (brs, 1H, NH), 7.25 (d, J = 7.3 Hz, 2H, Ar), 6.78 (d,
    J = 7.3 Hz, 2H, Ar), 6.25 (d, J = 8.1 Hz, 1H, NH), 4.10
    (hex, J = 8.1 Hz, 1H, CH), 3.66 (s, 3H, OCH3), 2.23-2.09
    (m, 2H, CH2), 1.90-1.71 (m, 2H, CH2), 1.64-1.51 (m, 2H,
    CH2)",
  "c_nmr": "154.8, 154.4, 134.0, 119.9, 114.3, 55.5, 45.0, 31.8, 14.8",
  "smiles": "C0c1ccc(NC(=O)NC2CCC2)cc1",
  "inchikey": "NTJIYHYWVZYBEX-UHFFFAOYSA-N",
  "has_structure": true,
  "license": "CC-BY",
  "license_pool": "commercial"
}
```

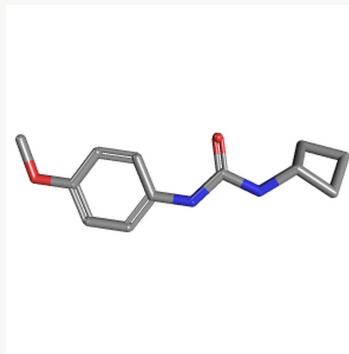


Figure S2: Reprint of main-text Listing 1 (listing1_irexp_record). IReXP JSON+molecule ($C_{12}H_{16}N_2O_2$; NTJIYHYWVZYBEX-UHFFFAOYSA-N). SMILES / InChIKey are resource fields, not solver payload. Not a Data Descriptor dump. Locked v3-R25 is the mechanism case (Figure S4), not this listing.

(151/149); the eligible IReXP corpus is 17.5% simple / 82.5% complex. Reweighting lives on validate-clean $n=295$ (§5) and is not a new metric.

3.2 Pre-registration pointers

These files live in the companion spectro-agent tree, not this postcard manuscript repo. Pointers are recorded so a reviewer can ask for the committed note rather than a reconstructed story.

- Locked $n=194$: main clean 134 + v3 40 + v2_ctrl 20. Spectrally validated main round plus two control arms. Sampler seeds for the original 140 / 40 draws were not captured (gap #4 in docs/SUBMISSION.md); the released roster is fixed and reusable but not exactly re-drawable.
- Expansion-106: pre-registered draw after the 194 lock, same irexp_resolved sampler. Five ^{13}C -overread flags R12, R22, R25, R82, R91 stay in the $n=300$ headline (ICLR lock 2026-09-17). Validate-clean $n=295$ is appendix / SI sensitivity.
- expand-500: docs/EXPANSION_PREREGISTRATION_500.md (seed 2026500; $n=230$; 115/115 simple/complex; exclusion of 375 prior InChIKey-14s including a local restore of the withheld +106 key). 0 collisions with any prior round. Six ^{13}C -overread flags known before any solve: R26, R31, R102, R105, R107, R138.

3.3 Stopping / pooling rules

- Partial expand-500 subsets are not a headline.
- score_pooled.py --expand-500 refuses a missing predictions2.jsonl or a withheld key.
- answers2.jsonl is not in either working tree (git ls-files empty for +106 and expand-500).
- Keys were restored only under /tmp/blind/_key/ for the 2026-09-20 score pass, then deleted.



Figure S3: Reprint of the appendix diagnostic wall (fig_wall_diagnostic): 58 verified / 7 mis-ranked / 129 never proposed on the locked $n=194$ slice. Same integers as Figure S1. No expansion-106 or expand-500 wall exists.

- validate_benchmark.py was not re-run on the night pass (would rewrite committed clean_qids.json). The existing 224-qid snapshot is used.
- collect_round.py was not run; 230 raw/single_R*.json files were left untouched.
- forward_verify_main.py score was not run on expand-500 (§8).
- -ci / bootstrap was not run on expand-500. Do not invent CIs.

3.4 What the solver may do

RDKit formula / parse check only. No web search. No retrieved spectrum. No thinking tier on the scored $n=300$ headline. The expand-500 deposits used a thinking-tier consumer arm and are labelled as such wherever they appear — they are not interchangeable with the no-thinking headline protocol.

A deposit is JSONL of {qid, candidates: [SMILES,...]} under a stated budget. Headline budget: ≤ 3 ranked SMILES per qid. Self-ranking vs forward-verified re-ranking must be labelled; they are not interchangeable.

4 Scoring contract (equations used in the main text)

These are the same identities as main-text Eqs. (1)–(7). They are restated so the SI is readable alone. No new metric is introduced.

A candidate $\hat{y}$ matches gold y^* at constitution iff

$$s(\hat{y}, y^*) = \mathbf{1}[\text{IK}_{14}(\hat{y}) = \text{IK}_{14}(y^*)]. \quad (1)$$

IK_{14} is the first 14 characters of the RDKit InChIKey (connectivity layer). Full stereo-sensitive InChIKey is reported only on validate-clean $n=295$ (§5).

With ranked pool $\mathcal{C}_i = (c_{i1}, \dots, c_{ik})$, $k \leq 3$,

$$\text{top-1} = \frac{1}{n} \sum_i s(c_{i1}, y_i^*), \quad (2)$$

$$\text{recall}@k = \frac{1}{n} \sum_i \mathbf{1}[\exists j \leq k : s(c_{ij}, y_i^*) = 1], \quad (3)$$

$$\text{prec.}|\text{recall} = \frac{\sum_i s(c_{i1}, y_i^*)}{\sum_i \mathbf{1}[y_i^* \in \mathcal{C}_i]}. \quad (4)$$

These are the three reportable numbers. They factorise as

$$\text{top-1} = (\text{recall}@k) \times (\text{prec.}|\text{recall}) \quad (5)$$

when the same pool is used for both stages — the rates still have different denominators and must not be differenced. Self-ranking $\text{prec.}|\text{recall}$ on $n=300$ is 118/133 (88.7%). That is not forward-verify precision.

Forward-verify re-ranks by symmetric chamfer on ^{13}C peak sets $P, Q \subset \mathbb{R}$

$$d(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} |p - q| + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} |q - p|, \quad (6)$$

where $\hat{P}(c)$ is predicted blind to the observed list P_{obs} . Instrumented only on the locked $n=194$ (58/65). Expansion-106: not run. expand-500: incomplete coverage, no score (§8).

A pool estimator on an index set $\mathcal{S}$ is

$$\hat{p}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} s(c_{i1}, y_i^*). \quad (7)$$

The ICLR headline uses $\mathcal{S} = \{1, \dots, 300\}$ (194 locked + all 106 Opus, flags included). Exploratory $\mathcal{S}$ of size 524 / 519 / 530 (§7) do not replace that headline. expand-500 is a separate $\mathcal{S}$ of size 230 (or clean 224) and is reported on its own row.

Confidence intervals on $n=300$ / $n=295$ / $n=194$ are bootstrap 95% (score_main.boot, seed 0, 2000 resamples). Paired model differences use McNemar with Holm correction. No CIs are reported for expand-500 or for the exploratory pools.

5 Locked headline and expansion-106

Source: docs/POOLED_HEADLINE_2026-09-16.md (python scripts/score_pooled.py on spectro-agent). Headline is $n=300$, flags included.

Table 2: Locked generation slices. Headline is $n=300$.

set	n	top-1	recall@3	notes
locked 194	194	55/194 (28.4%) [22–35]	65/194 (33.5%) [27–40]	instrumented fverify
locked simple	98	47/98 (48.0%)	53/98 (54%)	
locked complex	96	8/96 (8.3%)	12/96 (12%)	
expansion-106 all	106	63/106 (59.4%) [50–69]	68/106 (64.2%) [56–73]	enters headline
+106 simple	53	41/53 (77.4%)	—	
+106 complex	53	22/53 (41.5%)	—	
expansion-106 clean	101	61/101 (60.4%) [50–69]	65/101 (64.4%) [54–73]	appendix $n=295$
clean simple	49	40/49 (81.6%)	41/49 (84%)	
clean complex	52	21/52 (40.4%)	24/52 (46%)	
headline 194+all 106	300	118/300 (39.3%) [34–45]	133/300 (44.3%) [39–50]	flags in
headline simple	151	88/151 (58.3%)	96/151 (64%)	
headline complex	149	30/149 (20.1%)	37/149 (25%)	
validate-clean 194+101	295	116/295 (39.3%) [34–45]	130/295 (44.1%) [39–49]	appendix
clean-pool simple	147	87/147 (59.2%) [51–67]	94/147 (63.9%) [56–71]	
clean-pool complex	148	29/148 (19.6%) [14–26]	36/148 (24.3%) [17–31]	

Self-ranking precision | recall on the generation pool: 118/133 (88.7%) on $n=300$; 116/130 (89.2%) on $n=295$ — not forward-verify precision.

5.1 Five flagged +106 compounds

Table 3: +106 ^{13}C -overread flags. Included in $n=300$; out of $n=295$.

item	value
qids	R12, R22, R25, R82, R91
reason	^{13}C -overread (scripts/validate_benchmark.py)
snapshot	data/benchmark_expand/clean_qids.json
when flagged	before subset totals (stopping rule)
in paper headline?	yes ($n=300$, not $n=295$)
effect vs excluding	+2 top-1, +3 recall $\rightarrow$ 118/300 and 133/300
one-decimal top-1	unchanged (both 39.3%)

5.2 Validate-clean extras (not the headline)

These were scored on $n=295$ only. They were not re-run on the $n=300$ roster and must not be quoted as $n=300$ figures.

Table 4: Validate-clean $n=295$ extras. Same scorer; not the headline.

metric	overall	simple	complex
scaffold-level (best Tanimoto ≥ 0.45)	64%	80%	48%
mean best Tanimoto	0.66	0.79	0.53
full InChIKey (stereo) top-1	93/295 (31.5%) [26–37]	—	—
full InChIKey recall	108/295 (36.6%) [31–42]	—	—
corpus-reweighted top-1	26.5% [21–32]	—	—
corpus-reweighted recall	31.3% [25–37]	—	—
locked-194 stereo top-1	41/194 (21.1%)	—	—

Table 5: Size bins on validate-clean $n=295$.

heavy atoms	n	top-1	recall
≤ 15	53	36/53 (67.9%)	39/53 (73.6%)
16–25	152	65/152 (42.8%)	75/152 (49.3%)
> 25	90	15/90 (16.7%)	16/90 (17.8%)

On the locked 194, of 137 analysable top-1 misses, 76.6% are constitutional isomers of the true structure; only 22.6% share the true Murcko scaffold. Named-ring annotation on that slice: 10/194 author assignments name a ring system, solved 1/10 vs 54/184 for the rest. Neither analysis was re-run on the expansion. Fable expansion: 68/106 deposits; not scored; never pooled. Expansion-106 fverify: not run.

6 expand-500 generation (NIGHT_POOL)

Source: spectro-agent data/benchmark_expand_500/NIGHT_POOL_2026-09-20.md on cursor/expand-bench-500-b78b (copied to docs/NIGHT_POOL_2026-09-20.md). Key restored only for score2,

then deleted. answers2.jsonl is not in the tree.

Table 6: expand-500 generation (score2, InChIKey-14). No CIs.

set	n	top-1	recall@3
all deposited Opus	230	129/230 (56.1%)	138/230 (60.0%)
simple	115	82/115 (71%)	87/115 (76%)
complex	115	47/115 (41%)	51/115 (44%)
validate-clean	224	127/224 (56.7%)	135/224 (60.3%)
simple	110	81/110 (74%)	85/110 (77%)
complex	114	46/114 (40%)	50/114 (44%)

Mean best Tanimoto (score2 footer, all 230): simple 0.863; complex 0.645. Flagged qids (in 230, out of 224): R26, R31, R102, R105, R107, R138. Deposits: 230/230; predictions2.jsonl 230 lines, 690 candidates. Unique (formula, IR, ¹³C) vs irexp_resolved.jsonl.gz: 230/230; 0 prior-round InChIKey-14 collisions. Protocol: thinking-tier consumer arm — not interchangeable with the no-thinking $n=300$ headline.

Verbatim score2 footer (do not polish the integers):

overall recovered (top-3): 138/230 (60%) top-1 exact: 129/230 (56%)

simple : recovered 87/115 (76%) top1 82/115 (71%) meanBestTani 0.863

complex : recovered 51/115 (44%) top1 47/115 (41%) meanBestTani 0.645

7 Pool options (generation only)

The estimator is Eq. 7. Only the first row is the paper headline.

Table 7: Pool options. Headline is the first row. No CIs. No fverify.

n	composition	role	top-1	recall
300	194+all 106	paper headline	118/300 (39.3%)	133/300 (44.3%)
524	194+all 106+clean 224	largest clean-on-this-round additive	245/524 (46.8%)	268/524 (51.1%)
519	194+clean 101+clean 224	what score_pooled.py -expand-500 prints	243/519 (46.8%)	265/519 (51.1%)
530	194+all 106+all 230	all deposited, including 6 flags	247/530 (46.6%)	271/530 (51.1%)
295	194+clean 101	older lock; appendix sensitivity	116/295 (39.3%)	130/295 (44.1%)

Strata on the exploratory pools (same InChIKey-14 rule; not headline):

Table 8: Exploratory-pool strata. Copied from NIGHT_POOL extra-pool script.

pool	simple n	simple top-1 / recall	complex n	complex top-1 / recall
300	151	88/151 (58%) / 96/151 (64%)	149	30/149 (20%) / 37/149 (25%)
524	261	169/261 (65%) / 181/261 (69%)	263	76/263 (29%) / 87/263 (33%)
519	257	168/257 (65%) / 179/257 (70%)	262	75/262 (29%) / 86/262 (33%)
530	266	170/266 (64%) / 183/266 (69%)	264	77/264 (29%) / 88/264 (33%)

Do not treat 519, 524, or 530 as a replacement headline. Do not draw a pooled wall from these rows. Do not attach CIs. The official score_pooled.py -expand -expand-500 path prints $n=519$ because it clean-filters both expansions; the paper still leads with $n=300$ (flags in on +106).

8 Forward-verify coverage (honest gaps)

8.1 Locked $n=194$ — complete

Source: data/diagnosis.json / scripts/forward_verify_all.py.

Table 9: Instrumented forward-verify. This is the only complete fverify arm.

	60-compound arm	locked slice ($n=194$)
generation recall	19/60 (32%)	65/194 (34%)
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58/194 (30%)
precision recall — fverify	16/19 (84%)	58/65 (89%)
multi-candidate only — fverify	10/13 (77%)	30/37 (81%)
verified / mis-ranked / never proposed	—	58 / 7 / 129
derangement floor on $n=37$	—	54.0%; fverify +27.1 points
McNemar vs self-rank	—	$p=0.55$ (7 gained, 4 lost)

Main-text Figure 1 includes this wall (reprinted as Figure S3). It is not a pooled $n=300$ wall.

8.2 Expansion-106 — not run

No data/fverify_expand/ bundle. No precision number. Self-ranking 118/133 on $n=300$ is not a substitute.

8.3 expand-500 — coverage only, no score

Inventory docs/FVERIFY_EXPAND500_INVENTORY_2026-09-20.md. Blind ^{13}C campaign over 681 unique candidate SMILES prepared by scripts/forward_verify_expand500_keyless.py. is_true is still absent. forward_verify_main.py score was not run.

Table 10: expand-500 fverify coverage only. Not a score.

item	count
prep unique SMILES	681
fbatch files	41/41
raw/f*.json present	31/41
unique SMILES with a numeric ^{13}C list	516/681 (75.77%)
unique SMILES still missing	165
present batches (complete vs fbatch)	f1–f15, f21–f30, f36–f41
missing batches	f16–f20, f31–f35
present SMILES by block	f1–15: 255; f21–25: 85; f26–30: 80; f36–41: 96
missing SMILES by block	f16–20: 85; f31–35: 80
qids with every candidate covered	129/230
qids with at least one candidate covered	230/230
forward_verify_main.py score	not run
verification precision	does not exist

Sibling packing in the keyless prep put a qid’s three candidates in different batches, so a hole in f16–f20 / f31–f35 still leaves every qid with at least one covered SMILES. That is not a reason to quote precision: a complete-qid subset of 129 would still be an uncommitted, post-hoc slice.

HOSE-style lookup on the locked $n=194$ candidate sets ties self-ranking; a modest GNN reaches 59/65 (91%) vs the LLM verifier’s 58/65 (89%) — directional, not a new headline. Deranging observed ^{13}C pairings collapses precision ($p=0.001$).

9 Failure and success case panels

The three locked-slice modes in the main appendix, plus extra panels that use only already-reported numbers. IDs below are locked-round labels (expansion qids reuse R22/R25). No SMILES are invented for R06/R26; only v3-R25 molecules are drawn (Figure S4).

Table 11: Three-mode case grid (disclosed facts only). Same integers as main-text Table tab:cases.

mode	id / formula	disclosed facts
Proposal miss	main-R06 / $\text{C}_{29}\text{H}_{30}\text{FNO}_2$	True acridinedione never in the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused. This is the 129/194 wall.
Verify-save	v3-R25 / $\text{C}_{10}\text{H}_{14}\text{N}_2\text{O}$	True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm). IR 1679 cm^{-1} . Figure S4. Not Listing 1.
Verify-fail	v3-R26 / $\text{C}_{14}\text{H}_{11}\text{N}_3\text{O}_3$	True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm^{-1} .

9.1 Proposal miss — main-R06 / $\text{C}_{29}\text{H}_{30}\text{FNO}_2$

True acridinedione never enters the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused. This is the 129/194 wall: re-ranking cannot recover a structure that was never proposed (Eq. 5).

9.2 Verify-save — v3-R25 / $\text{C}_{10}\text{H}_{14}\text{N}_2\text{O}$

True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm, Eq. 6). IR 1679 cm^{-1} . Main-text Figure 5; SI Figure S4.

9.3 Verify-fail — v3-R26 / $\text{C}_{14}\text{H}_{11}\text{N}_3\text{O}_3$

True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm^{-1} . Ranking can fail when predicted ^{13}C lists are nearly tied. This is why we do not claim an accuracy advance from forward-verify (McNemar $p=0.55$; 7 gained, 4 lost on $n=194$).

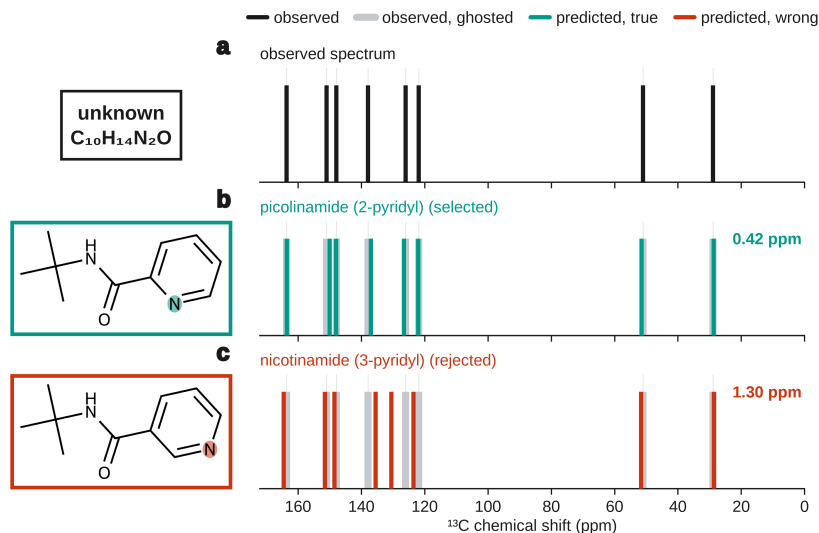


Figure S4: Locked v3-R25 verify-save (main-text Figure 5). Self-rank prefers 3-pyridyl; forward ^{13}C recovers 2-pyridyl (chamfer 0.42 vs 1.30 ppm). A verify-save, not a recall win. Not Listing 1.

9.4 Listing 1 is the IRExp record plate

Main-text Listing 1 is listing1_irexp_record ($\text{C}_{12}\text{H}_{16}\text{N}_2\text{O}_2$ urea; InChIKey NTJIYHYWVZYBEX-UHFFFAOYSA-N). SMILES / InChIKey / licence fields are resource metadata. Bench solvers see only formula and printed peak lists. Gold constitution is scored only by Eq. 1. Locked v3-R25 remains the mechanism worked case (main-text Figure 5), not Listing 1.

The plate is reprinted as Figure S2.

9.5 Extra panels (already-reported numbers)

Formula-only collapse (control, $n=60$). Formula only: 3/60 (5%) top-1 and 3/60 (5%) recall. Formula + IR + ^1H + ^{13}C : 14/60 (23%) / 19/60 (32%). Outcomes perfectly nested (McNemar $p=0.001$). The model is reading the peak lists, not only the formula.

Generate-wide (60-compound arm). Ten solver agents, up to six candidates each, pooled then re-ranked: recall 32% $\rightarrow$ 42%, top-1 23% $\rightarrow$ 30% (McNemar $p=0.34$); verification precision falls 84% $\rightarrow$ 72%. The training-free ceiling remains recall-limited. This is a success-for-recall panel, not an accuracy claim.

IRExp-fine-tuned generator (locked slice). Pooled with Claude: recall 33.5% $\rightarrow$ 54.1%, top-1 28.4% $\rightarrow$ 35.1% (McNemar $p=0.015$). The wall is a property of training-free elicitation, not of the task. Training is the one intervention already seen to move $P(\text{true} \in \text{pool})$.

Battery-electrolyte subset. $n=46$ literature electrolyte chemistry (not operando): 26% top-1, recall-bound — same regime as the locked slice.

Four-model Claude ladder ($n=24$). Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable is capability-sensitive but underpowered for adjacent ranks, with a disclosed protocol asymmetry. Not a headline.

Expansion difficulty gap. +106 is easier for this solver than the locked 194 (59.4% vs 28.4% top-1). That gap is a result, not a reason to hide the 106 or to keep $n=194$ as the headline. Pooling to $n=300$ does not erase it (main-text Table 2).

10 Prompt skeletons

10.1 Headline solver-facing skeleton

Wrapping instruction text was not captured on the headline Claude arm (Limitations (i); docs/-SUBMISSION.md gap #1). Only the per-compound four-field payload is released. The skeleton below is the contract, not a recovered verbatim system prompt.

You are given only:
 formula (as from HRMS)
 ir_bands_cm-1 (author-transcribed experimental list)
 h_nmr, c_nmr (printed shift lists)
 Return ≤ 3 ranked SMILES.
 No name, no SMILES hint, no scaffold, no retrieved spectrum.
 Check molecular formula with RDKit before emitting a candidate.

Listing 1 payload (fields only; ^1H abbreviated with ellipsis in the main float; gold held out):

```
{
  "formula": "C12H16N2O2",
  "ir_bands_cm-1": [3337.0, 2969.0, 1633.0, 1556.0,
                    1505.0, 1240.0, 1181.0, 1071.0, 826.0],
  "h_nmr": "8.07 (brs, 1H, NH), 7.25 (d, J = 7.3 Hz, 2H, Ar), ...",
  "c_nmr": "154.8, 154.4, 134.0, 119.9, 114.3, 55.5, 45.0, 31.8, 14.8"
}
```

10.2 Forward-verify skeleton

For each candidate SMILES, predict a ^{13}C peak list without seeing the observed list. Re-rank by $d(\hat{P}(c), P_{\text{obs}})$ (Eq. 6). The instruction dispatched to the forward-prediction sub-agents was likewise not captured; only the candidate SMILES and the resulting numeric lists are on disk for $n=194$.

10.3 Cross-vendor harness

The verbatim prompts in the (unreleased-here) ESI S2 of the spectro-agent tree are the cross-vendor harness prompts, not the missing headline Claude wrapper. External submissions must document model version, prompt, and tool access (docs/LEADERBOARD.md).

10.4 Deposit schema

```
{"qid": "R01", "candidates": ["SMILES_rank1", "SMILES_rank2", "SMILES_rank3"]}
```

Score locally (spectro-agent):

```
python scripts/score_submission.py --predictions my_run.jsonl --name "YourModel-1.0"
python scripts/score_pooled.py      # n=300 headline = 194+all 106
python scripts/score_pooled.py --stereo # 93/295 on validate-clean
python scripts/score_main.py        # locked n=194
python scripts/forward_verify_all.py # n=194 instrumented slice
```

11 Hyperparameters, hardware, and model cards

11.1 Headline model card

Table 12: Headline solver. Gaps are recorded, not filled.

item	value
headline solver	Claude Opus, consumer subscription
thinking tier (headline)	none
snapshot / seed / temperature / top_p / top_k / max_tokens	not exposed; not recorded (SUBMISSION gaps #2-#3)
candidate budget	≤ 3 SMILES / qid
tool access	RDKit formula / parse check only; no web search
fine-tuning	none on the headline arm
paid API	no, for the core protocol
hardware	consumer harness; no cluster job for inference
scoring hardware	commodity CPU; RDKit InChIKey-14
cross-vendor arm	$n=60$; Grok 4.6, Gemini 3.7 Flash, GPT-5.6 Sol
Claude mean candidates on that arm	2.20 (others 3.00) — recall rankings approximate
expand-500 solver	Claude Opus thinking-tier (declared; not headline)
Fable +106	68/106 deposits; not scored; not pooled
scoring scripts	scripts/score_pooled.py, score_main.py, bench-mark_v2.py score2

11.2 Cross-vendor card ($n=60$)

Table 13: Cross-vendor decomposition. Recall and precision use different denominators.

model	generation recall	prec. recall	multi-candidate only
Claude Opus	19/60 = 32%	16/19 = 84%	10/13 = 77%
Grok 4.6	32/60 = 53%	20/32 = 62%	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50%	22/30 = 73%	22/30 = 73%
GPT-5.6 Sol	25/60 = 42%	17/25 = 68%	16/24 = 67%

Verification precision exceeds generation recall in every arm. Bootstrap separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok’s gap (+9.2) is directional. A clean-clone control for Grok bounds key leakage (McNemar $p=0.39$). Reasoning-effort tier per vendor arm was not recorded (gap #2).

11.3 Literature decomposition (copied, not re-derived)

Any system reporting top-1 = a and top- k = b implies recall $\geq b$ and conditional precision $\leq a/b$. Script: scripts/literature_decomposition.py.

Three groups changed only the data, and recall carried 68% (NMR-Solver simulated $\rightarrow$ real), 70% (SpecX random $\rightarrow$ scaffold) and 83% (Espejo education $\rightarrow$ industrial) of each collapse. The solver-alone precision column on $n=300$ is self-ranking (118/133); the forward-verify row is the instrumented $n=194$ slice (58/65).

Table 14: Selected literature decomposition rows (as in main-text Table 6).

system (<i>n</i>)	top-1	top- <i>k</i> (<i>k</i>)	recall	prec.
this work, solver alone (300)	39.3%	44.3% (3)	44.3%	88.7%
this work + forward-verify (194)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver, literature (450)	52.9%	67.3% (10)	$\geq 67.3\%$	$\leq 78.6\%$
NMRAgent, literature (450)	61.6%	70.0% (10)	$\geq 70.0\%$	$\leq 88.0\%$
Espejo Morales, education (236)	80.9%	90.0% (5)	$\geq 90.0\%$	$\leq 89.9\%$
Espejo Morales, industrial (34)	20.6%	29.1% (5)	$\geq 29.1\%$	$\leq 70.9\%$
Alberts, 6–13 atoms, NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
SpecX, scaffold split, simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%

12 Two-paper split and anonymity

IRSpectra-Bench (this ICLR paper) is the blind peak-list benchmark and the propose $\gg$ verify diagnosis. IRexp is a companion Scientific Data Data Descriptor (in preparation): 121,233 records; 43,060 structure-linked; 33,201 full IR+ ^1H + ^{13}C +structure quadruples. This SI cites that resource and does not re-present Methods / Data Records / licence-pool narrative.

Double-blind review copy: <https://anonymous.4open.science/r/peaklist-corpus-review-10C4/>. Named Hugging Face / GitHub hosting is restored at camera-ready. Do not put ilkhamfy/* URLs in the PDF. Fine-tunes must use data/train_no_bench.jsonl.gz so benchmark InChIKeys are withheld.

Listing 1 is the only IRexp-style JSON+molecule plate in the ICLR PDF. It is a pointer, not a second paper.

13 Reproducibility checklist

ICLR-style checklist, filled honestly. “N/A” is used only when the item does not apply; gaps are named.

1. Code / data / frozen deposits. Frozen questions / predictions for $n=194$ and +106 live in spectro-agent. This postcard repo is the ICLR manuscript. Camera-ready will name the archival host.
2. Headline regeneration. python scripts/score_pooled.py regenerates 118/300 and 133/300 when the withheld +106 key is restored locally.
3. Locked-slice fverify. python scripts/forward_verify_all.py regenerates 58/7/129 on $n=194$ only.
4. expand-500 score2. Restore /tmp/blind/_key/benchmark_expand_500.answers2.jsonl.withheld, then benchmark_v2.py score2 -outdir data/benchmark_expand_500. Re-withhold the key. Confirm git ls-files data/benchmark_expand_500/answers2.jsonl is empty.
5. Do not commit answers2.jsonl for +106 or expand-500.
6. Do not invent CIs for expand-500; -ci was not run.
7. Do not run or quote forward_verify_main.py score on expand-500 until f16–f20 and f31–f35 exist.
8. Double-blind. Companion bib is Anonymous; dataset URL in the PDF is only the anonymous.4open review copy.
9. Fine-tunes must use data/train_no_bench.jsonl.gz.
10. IRexp construction, licences, and transcription audit belong in the Scientific Data companion — not in this SI beyond Listing 1.
11. Inference is not bit-exact. No pinned snapshot, temperature, or seed (Limitations (i)). Distributional agreement of frozen deposits is the intended claim.
12. Uncaptured artefacts (cannot be recovered): headline Claude wrapper text; dated model snap-

shots; decoding parameters; main-round sampler seeds; closed-book audit transcripts (docs/SUBMISSION.md gaps #1–#5).

13. Deferred arms. Expert-chemist audit not run. Leave-one-modality-out abandoned. Parser extraction-recall audit deferred to Sci. Data. SpectraLLM / IR-Agent / Spectro / NMIRacle / Alberts / CASE not scored on this bench.
14. Figure rebuild limits. fig_wall_diagnostic / Fig. 1 wall reads $n=194$ diagnosis.json. No expansion fverify bundle exists, so a pooled wall cannot be rebuilt. Size / recency / miss-isomer / named-ring analyses were never written against the expansion roster.
15. This manuscript compile. python3 scripts/build_pdf.py (main); python3 scripts/build_si.py (this SI). Engine: tectonic 0.17.0 or pdfLaTeX+BibTeX / XeLaTeX+BibTeX.

13.1 What a reviewer can re-check without the withheld keys

- Every a/b ($c\%$) in the main text and this SI is internally consistent with the integers in docs/POOLED_HEADLINE_2026-09-16.md and docs/NIGHT_POOL_2026-09-20.md.
- Fig. 1 caption, Table 2, Limitations (vi), and Appendix FAQ agree: wall = $n=194$; headline = $n=300$ generation.
- expand-500 rows always carry “no CIs / no fverify precision / thinking-tier / not the headline”.

14 Reviewer FAQ (SI copy)

These answers restate main-text Limitations and Appendix B. They add no new metrics.

Why is top-1 39% if the wall figure looks worse? Headline generation is $n=300$ (118/300 top-1; 133/300 recall). Figure 1 includes the locked $n=194$ forward-verified wall (58/7/129). Do not read the wall as a pooled $n=300$ diagnosis.

Is the expansion hidden in the pool? No. Expansion top-1 is 59% vs 28% on the locked 194. That gap is a result.

Where is forward-verify on the 106? Not run. Conditional 89% (58/65) is $n=194$ only.

Where is the $n\approx 500$ top-1? This SI / main appendix, not the headline: 129/230 (56.1%) top-1, 138/230 recall. Thinking-tier arm; no CIs; no fverify precision. Headline stays $n=300$ (Eq. 7).

Consumer harness / no snapshot / no thinking tier? Correct and intentional for the scored headline. Scoring of frozen deposits is reproducible; inference is not bit-exact.

Why no SpectraLLM / IR-Agent / CASE SOTA table? Different objects. None is scored on this bench. The paper is a stage diagnosis, not a method race.

Is IRexp a second ICLR paper? No. IRexp is a companion resource pointer (Listing 1; §12).

Should the headline move to $n=524$? No. Eq. 7 uses $\mathcal{S}$ of 300. 524 / 519 / 530 are exploratory and mix a thinking-tier arm with the no-thinking headline protocol.

What this SI is not

A Data Descriptor. A pooled wall. An expand-500 fverify result. A recommendation to switch the ICLR headline to $n=524$. A source of invented CIs or new metrics.